

Unit 03: Knowledge Representation and Reasoning

3.1 Propositional and Predicate Logic

Propositional and Predicate Logic are formal logical systems used to represent knowledge in a precise, structured, and unambiguous manner so that machines can perform reasoning.

- **Propositional Logic** represents knowledge using whole statements (propositions) that are either true or false.
- **Predicate Logic (First-Order Logic)** extends propositional logic by allowing representation of objects, properties, relationships, and quantifiers, making it more expressive and suitable for real-world problems.

3.1.1 Syntax, Semantics, and Inference (Forward Chaining, Backward Chaining)

A. Syntax

Syntax refers to the formal rules that determine how symbols, variables, predicates, and logical connectives can be combined to form valid logical expressions.

In simple words, syntax defines the **structure or grammar** of logical sentences.

In Propositional Logic:

- Propositions: P, Q, R
- Connectives:
 - $\neg$ (NOT)
 - $\wedge$ (AND)
 - $\vee$ (OR)
 - $\rightarrow$ (IMPLIES)
 - $\leftrightarrow$ (IFF)

Example:

$P \rightarrow Q$

(Student studies $\rightarrow$ Student passes)

A sentence is syntactically correct if it follows formation rules.

In Predicate Logic:

Syntax includes:

- Constants: Ram, Book
- Variables: x, y
- Predicates: Student(x), Pass(x)

- Quantifiers:
 - $\forall$ (for all)
 - $\exists$ (there exists)

Example:

$\forall x (\text{Student}(x) \rightarrow \text{Pass}(x))$

Predicate logic is more expressive than propositional logic.

Reference: Logic for Computer Science

B. Semantics

Semantics refers to the meaning and interpretation of logical sentences. It determines under what conditions a statement is true or false.

It determines:

- Whether a statement is TRUE or FALSE
- Under what interpretation (model) it becomes true

Example:

If

$P = \text{True}$

$Q = \text{False}$

Then

$P \rightarrow Q = \text{False}$

A model is an assignment of truth values that makes a sentence true.

C. Inference

Inference is the logical process of deriving new conclusions or knowledge from existing facts, premises, or previously known statements using valid reasoning rules.

In other words, inference allows a system to generate new information that was not explicitly stated but logically follows from the given knowledge.

Important inference rule:

Modus Ponens:

1. $P \rightarrow Q$
2. P
3. Therefore Q

Forward Chaining

Forward chaining is a data-driven inference technique in which reasoning starts from known facts and repeatedly applies logical rules to derive new facts until a desired conclusion or goal is reached.

In this method, the system moves **forward from available data toward conclusions**.

Steps:

1. Start with known facts
2. Apply rules
3. Generate new facts
4. Continue until goal reached

Example:

Fact: Student completed assignment

Rule: If assignment completed → eligible for exam

Conclusion: Student eligible for exam

Used in:

- Expert systems
- Monitoring systems

Backward Chaining

Backward chaining is a goal-driven inference technique in which reasoning starts from a desired conclusion (goal) and works backward to determine whether the known facts support that goal.

In this method, the system begins with what it wants to prove and then searches for rules and facts that can justify it.

Steps:

1. Start with goal
2. Find rule that produces goal
3. Check if conditions are true
4. Continue backward

Example:

Goal: Student passes

Check rule: If studies → passes

Check if student studies

Used in:

- Diagnostic systems
- Intelligent tutoring systems

Comparison:

Forward Chaining	Backward Chaining
Data-driven	Goal-driven
Starts from facts	Starts from goal
Generates many facts	Focused search

3.2 Rule-based Systems and Statistical Reasoning

3.2.1 Probability, Bayes' Theorem, and Belief Networks

A. Rule-Based Systems

A rule-based system is an artificial intelligence system that represents knowledge in the form of IF–THEN rules and applies logical reasoning to make decisions or derive conclusions.

It operates by matching facts in a knowledge base with predefined rules and generating conclusions through an inference mechanism.

Structure:

IF condition THEN action

Example:

IF marks ≥ 90 THEN grade = A

Components:

1. Knowledge Base (facts + rules)
2. Inference Engine
3. User Interface

Advantages:

- Easy to understand
- Transparent reasoning

Disadvantages:

- Difficult to manage large rule sets
- Cannot easily handle uncertainty

B. Probability

Probability is a mathematical measure of uncertainty that quantifies the likelihood of an event occurring. It assigns a numerical value between **0 and 1**, where:

- 0 indicates the event is impossible.
- 1 indicates the event is certain.

Range:

$$0 \leq P(A) \leq 1$$

Formula:

$$P(A) = \text{Favorable outcomes} / \text{Total outcomes}$$

Conditional Probability:

$$P(A|B) = P(A \cap B) / P(B)$$

Used in:

- Student performance prediction
- Risk analysis

C. Bayes' Theorem

Bayes' Theorem is a mathematical formula used to update the probability of a hypothesis after observing new evidence. It allows us to revise our initial belief (prior probability) in light of additional information.

Formula:

$$P(A|B) = [P(B|A) \times P(A)] / P(B)$$

Where:

- $P(A)$ = Prior probability
- $P(B|A)$ = Likelihood
- $P(A|B)$ = Posterior probability

Example:

A = Student understands topic

B = Student answers correctly

After correct answer, belief about understanding increases.

D. Belief Networks (Bayesian Networks)

A **belief network**, also called a **Bayesian network**, is a graphical model that represents **probabilistic relationships among a set of variables**. It uses a directed acyclic graph (DAG) where:

- **Nodes** represent random variables.
- **Edges** represent conditional dependencies between variables.

Structure:

- Nodes $\rightarrow$ variables
- Directed edges $\rightarrow$ dependencies
- Conditional Probability Tables (CPT)

Example:

Study Habits → Understanding → Exam Result

Properties:

- Handles uncertainty
- Based on conditional independence
- Efficient reasoning

3.3 Applications in Education

AI-based reasoning is widely used in education systems.

3.3.1 Automated Grading and Lesson Planning

A. Automated Grading

Automated grading systems are AI-based systems that evaluate student answers automatically using predefined rules, algorithms, or machine learning techniques. These systems can assess both **objective** and **subjective** responses without human intervention.

Types:

1. Objective grading (MCQs)
2. Essay grading (NLP-based)
3. Programming code evaluation

Techniques used:

- Rule-based systems
- Statistical models
- Machine learning

Benefits:

- Fast feedback
- Consistency
- Scalable evaluation

Used by institutions like Educational Testing Service.

B. Lesson Planning

AI-based lesson planning refers to the use of intelligent systems to assist teachers in designing, organizing, and personalizing learning content based on student performance, learning needs, and curriculum requirements.

AI supports teachers by:

1. Analyzing student data

2. Identifying weak topics

3. Recommending personalized content

Uses:

- Bayesian networks
- Rule-based recommendation systems

Example:

IF student weak in algebra

THEN recommend basic algebra module